



Khulna University of Engineering & Technology (KUET)

Department of Computer Science and Engineering

Machine Learning Lab Report

*A Machine Learning Framework for Predicting and Ranking the Content Quality of
Bangladeshi University Websites*

Course Information

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1 Title

Main title. A Machine Learning Framework for Predicting and Ranking the Quality of Bangladeshi University Websites.

2 Project Scope

Existing tools for university website quality are rule-based: a human fixes the weights (e.g. 50% content / 40% performance / 10% load time) and the tool applies arithmetic. This project proposes a machine-learning framework that extracts website features automatically and learns to predict independently collected human quality ratings. The traditional weighted score is retained only as a rule-based baseline for comparison.

Objectives

- O1.** Develop a crawler and parser that produce a fixed feature vector for each university website.
- O2.** Construct a labelled dataset covering Bangladeshi university websites.
- O3.** Rank university websites using model-predicted continuous quality scores and compare their dimension-level performance.
- O4.** Generate actionable, SHAP-based explanations for the predicted quality scores.

Deliverables: crawler, raw and processed datasets, training and evaluation notebooks, model artefacts with a `predict(url)` entry point, ranking dashboard, and final report.

3 Literature Review

Previous studies on university website quality have applied automated web analysis, questionnaire-based evaluation, usability assessment, and web-quality models. The studies most closely related to the present work are summarised in Table 1. Particular attention is given to studies that evaluate university websites using automatically measurable content, usability, accessibility, performance, and technical attributes.

Table 1: Summary of the literature most relevant to the proposed university website evaluation framework

#	Reference	Focus / method	What we reuse
1	Rashida et al. (2021), <i>Computers</i> 10(5):57	Proposed a multi-method framework for evaluating university websites in Bangladesh. The automated tool uses Selenium and ChromeDriver for web crawling and evaluates three major criteria: content of information, website performance, and loading time. Under content of information, 25 key strings are searched, while 20 browser timing variables are collected for performance and loading-time analysis. A questionnaire-based statistical evaluation was also conducted to obtain users' opinions and compare the survey-based ranking with the automated ranking.	This is the principal baseline for the present work. We reuse and extend the idea of automated university website evaluation, landing-page content detection, browser-based scraping, technical performance measurement, and user-oriented validation. Several of our content attributes such as vision and mission, faculty information, notice board, academic information, library, contact information, and other university-specific resources originate from or extend the content criteria used in this study.
2	Saleh et al. (2022), <i>Indonesian Journal of Electrical Engineering and Computer Science</i> 25(1):511–520	Conducted a systematic literature review of university website quality evaluation. The study initially extracted 79 quality factors from previous research and, after comparison and filtration, consolidated them into six major factors: information quality, specific content, usability, web appearance, service interaction quality, and functionality. The study also reported the absence of a universally accepted specific model for university website quality evaluation.	We use the six-factor structure as a theoretical foundation for grouping and justifying the website attributes used in our dataset. It particularly supports the inclusion of information/content attributes, usability features, appearance features, interaction facilities, accessibility-related elements, and functional services available through university websites.

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Table 1 – continued from previous page

#	Reference	Focus / method	What we reuse
3	Mukanda, Mbuguah & Wabwoba (2022), <i>International Journal of Computer Trends and Technology</i> 70(2):1–9	Evaluated the usability of the top five Kenyan university websites for Webometrics ranking using automated web-analysis tools. The evaluated criteria included URL structure, website age, uptime, content management system, responsiveness, browser compatibility, colour contrast, domain authority, backlinks, website traffic, broken links, number of subdomains, website speed, indexed pages, website compression, SEO, website content utility, website design, and website accessibility. Tools including Nibbler, Silktide, Semrush, SEO Site Checkup, and Ahrefs were used for automated assessment.	This study directly supports the technical and usability portion of our feature set. From its evaluation criteria, our work adopts or adapts measurable attributes such as mobile responsiveness, colour contrast, broken links, page load speed, page compression, SEO-related information, and accessibility. Unlike this study, which evaluated only five universities, our dataset extends automated evaluation to a substantially larger and geographically diverse set of university websites.
4	Biyyapu et al. (2023), <i>Computers</i> 12(9):181	Applied machine-learning techniques to website quality classification using automatically extracted completeness-related attributes. The parser considers seven missing-resource or structural measures, including missing URLs, images, videos, PDFs, table columns, form fields, and self-referencing hyperlinks. An MLP architecture was compared with SVM and other classification approaches for predicting website-quality levels.	This work motivates the transition from a fixed rule-based score to a learned machine-learning model. We retain the idea of representing each website as a structured feature vector and comparing multiple machine-learning algorithms. However, our feature space is substantially broader because it includes content, navigation, ranking and recognition, events and media, technical performance, SEO, service interaction, and accessibility features rather than only resource completeness.
5	Ranking Web of Universities (Webometrics) methodology	Webometrics evaluates the web presence and visibility of higher-education institutions using externally observable web indicators. It demonstrates that university websites can be analysed quantitatively and that institutional web presence can provide useful information for comparative evaluation and ranking.	Webometrics provides an external point of comparison for examining the construct validity of the ranking produced by our proposed system. External ranking values are used for comparison and validation where appropriate rather than being used directly to determine the learned website-quality score.

4 Data Collection

4.1 Population and Frame

The dataset for this study was constructed from university websites collected from multiple geographical regions around the world. A total of **1,200 university websites** were collected by six members of the research team, with each member collecting data from approximately **200 universities**. The geographical distribution was divided among the team members to obtain coverage from North America, Europe, Asia, Latin America, and Africa.

Name	Roll	Area of Universities Collected
Nabil	<i>2107073</i>	North America
Abhijit	<i>2107083</i>	Western & Northern Europe
Afifa	<i>2107087</i>	Eastern & Southern Europe
Faisal	<i>2107061</i>	Asia Part 1
Mithu	<i>2107066</i>	Asia Part 2
Mahi	<i>2107076</i>	Latin America and Africa
Total	–	$6 \times 200 = 1,200$ Universities

4.2 Dataset Size by Granularity

The primary dataset contains **1,200 university websites**. For each university, attributes visible or verifiable from the university’s landing (home) page were collected. The attributes were designed to support both automated scraper implementation and downstream machine-learning feature encoding.

The attribute set was developed from three main sources:

- **P1:** Mukanda et al., “Usability Evaluation of Kenyan Universities Websites for Webometrics Ranking” (IJCTT, 2022).
- **P2:** Rashida et al., “Towards Developing a Framework to Analyze the Qualities of the University Websites” (Computers, 2021).
- **P3:** Saleh et al., “Systematic Literature Review on University Website Quality” (Indonesian Journal of Electrical Engineering and Computer Science, 2022). This study identifies six major website-quality factors: information quality, specific content, usability, web appearance, service interaction quality, and functionality.

4.3 University Landing-Page Attributes Used for Data Collection and Evaluation

The following attributes were used during data collection and subsequent university website evaluation. The attributes include Boolean presence/absence features, numerical counts, continuous measurements, dates, and other scraper-verifiable properties.

Table 2: Landing-Page Attributes Used for University Website Evaluation

#	Attribute	Data Type	Value Range / Format	Justification	Reference / Source
1. Header & Navigation					
1	Logo / institutional branding	Boolean	{0,1}	Institutional identity	P2
2	Primary navigation menu presence	Boolean	{0,1}	Core wayfinding, ease of use	P1; P2
3	Number of top-level nav items	Integer	[0–20]	Navigation complexity	P3 (easy to navigate)
4	Search bar	Boolean	{0,1}	Fast task completion	P1 (efficiency)
5	Language toggle (native/English)	Boolean	{0,1}	Multilingual accessibility	P2 (Q17); P3 (multi-language)
6	Breadcrumb / current-location cue	Boolean	{0,1}	Recognition over recall	P1 (heuristics)
2. Rankings & Recognition (new)					
7	QS World Ranking badge/mention	Boolean	{0,1}	Global prestige signal	New (practical addition)
8	QS World Ranking value (if shown)	Integer, nullable	[1–2000] or null	Comparable prestige metric	New
9	National/local ranking mention	Boolean	{0,1}	Local prestige signal	New; P1 (webometrics theme)
10	Webometrics rank value (if shown)	Integer, nullable	[1–30000] or null	Benchmark comparability	P1 (webometrics ranking)
11	Accreditation badges	Boolean	{0,1}	Credibility signal	New
12	Accreditation badge count	Integer	[0–10]	Credibility depth	New
13	Recent achievements / awards section	Boolean	{0,1}	Institutional reputation	New
14	Impact/highlight stats block (funding, jobs, patents)	Boolean	{0,1}	Quantified credibility	New (e.g. Stanford stats block)
15	Number of highlight stat items shown	Integer	[0–10]	Richness of proof points	New
3. Notices, Announcements & Updates					
16	Notice board presence	Boolean	{0,1}	Core recurring info need	P2 (key string 17; Q22)
17	Notice timestamp shown (date posted)	Boolean	{0,1}	Signals content freshness	P2 (Q22)
18	Most recent notice date	Date (ISO 8601), nullable	e.g. 2026-08-01 or null	Currency measurement	P2 (freshness)
20	News & events section	Boolean	{0,1}	Institutional activity visibility	P2 (key string 18)
21	Events / academic calendar link	Boolean	{0,1}	Time-sensitive planning info	P1; P2 (key string 25)

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#	Attribute	Data Type	Value Range / Format	Justification	Reference / Source
22	Admission announcements / deadlines	Boolean	{0,1}	High-frequency student query	P2 (key string 23)
4. Events & Media (new)					
23	Upcoming events section	Boolean	{0,1}	Engagement & currency	New (e.g. Oxford events block)
24	Number of upcoming events listed	Integer	[0-20]	Activity richness	New
25	Event images present	Boolean	{0,1}	Visual engagement	New; P3 (use of images)
26	Event captions / descriptions present	Boolean	{0,1}	Informativeness	New
27	Event date & time displayed	Boolean	{0,1}	Actionability	New
28	Forthcoming contests or competitions	Boolean	{0,1}	Student engagement signal	New
29	Video content on landing page	Boolean	{0,1}	Rich media engagement	P3 (proper use of multimedia)
30	Photo / image gallery	Boolean	{0,1}	Visual storytelling	P3 (appropriate use of images)
31	Social media feed embed	Boolean	{0,1}	Real-time engagement	P2 (online facilities theme)
5. Landing-Page Content & Information					
32	Vision & mission statement	Boolean	{0,1}	Institutional identity signal	P2 (key string 1)
33	About / overview blurb	Boolean	{0,1}	First-impression orientation	P2 (content of info)
34	Department list / quick links	Boolean	{0,1}	Primary navigation need	P2 (key string 10)
35	Faculty highlight or link	Boolean	{0,1}	Academic credibility cue	P2 (key string 3)
36	Research / publication highlight	Boolean	{0,1}	Academic reputation signal	P2 (key strings 6,8); P3
37	Academic programs listing	Boolean	{0,1}	Prospective-student decision info	P2 (key string 9)
38	Scholarship info / link	Boolean	{0,1}	Strong student interest area	P2 (key string 21; Q4=53%)
39	Library access / link	Boolean	{0,1}	Common student resource	P2 (key string 15)
40	Career / job portal link	Boolean	{0,1}	Student utility feature	P2 (key string 12)
41	Alumni section / link	Boolean	{0,1}	Institutional trust signal	P2 (key string 19)
42	FAQ link	Boolean	{0,1}	Self-service reduces friction	P2 (key string 24)

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#	Attribute	Data Type	Value Range / Format	Justification	Reference / Source
43	Contact info / webmail link	Boolean	{0,1}	Direct communication channel	P2 (key string 2)
44	Student portal link	Boolean	{0,1}	Core functional access	P3 (student portal)
45	Prospectus / brochure download	Boolean	{0,1}	Decision-support material	P3 (brochure)
46	Admissions policy / procedure link	Boolean	{0,1}	Decision-critical info	P3 (admissions, academic policies)
6. Footer Details					
47	Footer contact details	Boolean	{0,1}	Standard usability expectation	P2
48	Footer sitemap link	Boolean	{0,1}	Redundant navigation path	P1
49	Copyright / last-updated line	Boolean	{0,1}	Currency & credibility cue	P1
50	Social media links	Boolean	{0,1}	Engagement/communication channel	P2
51	Quick-links block	Boolean	{0,1}	Reduces clicks to key pages	P2
7. Visual Design					
53	Colour contrast ratio	Float	[1–21] (WCAG scale)	Readability & accessibility	P1 (Table VIII)
54	Banner / carousel content	Boolean	{0,1}	First-impression engagement	P2
57	Noticeable/prominent logo placement	Boolean	{0,1}	Brand recall	P3 (noticeable logo)
8. Service Interaction Quality (new, from P3)					
58	Live chat / chatbot presence	Boolean	{0,1}	Real-time support	P3 (responsiveness, additional services)
59	Feedback / contact form on landing page	Boolean	{0,1}	Two-way interaction	P3 (service interaction quality)
60	Trust/security seal displayed	Boolean	{0,1}	Perceived trust	P3 (trust, secure information)
61	Testimonials / student stories section	Boolean	{0,1}	Sense of community	P3 (sense of community)
9. Technical Performance					
62	Page load speed	Float	seconds, [0–15+]	Directly affects usability	P1 (Table XV); P2 (10% of score)
63	Mobile responsiveness score	Float (score)	[0–100]	Multi-device access need	P1 (Table VI)
65	HTTPS / SSL presence	Boolean	{0,1}	Security & user trust	P2 (Https criterion)

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#	Attribute	Data Type	Value Range / Format	Justification	Reference / Source
66	Broken links on landing page	Integer	[0–N] count	Reliability & credibility	P1 (Table XIII)
67	Page compression enabled (Gzip etc.)	Boolean	{0,1}	Faster, lighter load	P1 (Table XVI)
10. SEO & Metadata					
69	Page title & meta description present	Boolean	{0,1}	Search-engine visibility	P1 (SEO criteria)
70	Favicon presence	Boolean	{0,1}	Brand recognition in-browser	General convention
71	Sitemap / robots.txt reference	Boolean	{0,1}	Crawlability, indexing	P1 (SEO criteria)
11. Accessibility & Functionality					
72	Alt-text coverage on images	Float (%)	[0–100]	Screen-reader accessibility	P1 (accessibility score)
73	Accessible design across devices	Boolean	{0,1}	Inclusive access for all	P1 (accessibility performance)
74	Text-size / contrast accessibility toggle	Boolean	{0,1}	Inclusive design option	P3 (text-only option)
75	Bookmark / save-page facility	Boolean	{0,1}	Convenience functionality	P3 (bookmark facility)

4.4 Attribute Selection and Scraping Strategy

The attributes were restricted to information that is visible on the university landing page or is directly verifiable from the landing page. The majority of the attributes use Boolean values (0/1) to represent the presence or absence of a particular element. This representation is scraper-friendly and allows reliable DOM- and keyword-based detection.

Integer-valued attributes were used for measurable counts such as the number of navigation items, events, accreditation badges, highlight statistics, and broken links. Float-valued attributes were used where continuous measurements were required, such as page-load speed, colour contrast ratio, mobile responsiveness, and alt-text coverage.

Ranking attributes such as QS World Ranking and Webometrics ranking were recorded as nullable values because not every university displays such information on its landing page. The most recent notice date was stored using the ISO 8601 date format when such a date was available.

Two categories, *Rankings & Recognition* and *Events & Media*, were included based on practical observations from real university landing pages. Examples include ranking badges, accreditation information, achievement highlights, impact statistics, upcoming events, event images, videos, galleries, and similar landing-page elements.

The third literature source, P3, contributed landing-page-relevant attributes associated with its six website-quality factors: information quality, specific content, usability, web appearance, service interaction quality, and functionality. These include features such as student portal access, brochure/prospectus availability, live chat, trust or security seals, testimonials, bookmark facilities, and accessibility options.

4.5 Data Collection Protocol

For each university, the verified official university domain was used as the starting URL. The landing page was loaded and the selected attributes were extracted using automated scraping wherever possible.

Boolean attributes were encoded as:

$$x_i = \begin{cases} 1, & \text{if the corresponding attribute is detected,} \\ 0, & \text{otherwise.} \end{cases} \quad (1)$$

Count-based attributes were stored as integers, continuous measurements as floating-point values, and unavailable nullable measurements were stored as null values.

The scraper analysed HTML and DOM elements, hyperlinks, metadata, text content, images, media elements, accessibility properties, and technical response information according to the corresponding attribute. Where JavaScript-rendered content was required, the page was rendered using a browser-based scraping environment before attribute extraction.

During crawling, automated requests were handled responsibly by respecting `robots.txt` restrictions and avoiding form submission or other actions that modify the target website. The objective of the crawler was limited to collecting publicly available information required for university website evaluation.

4.6 Final Dataset Structure

After data collection, each university was represented by one feature vector. Since six attributes from the original 75-attribute specification were excluded from the present study, the final evaluation dataset contains **69 retained attributes per university**.

Thus, the primary dataset can be represented as:

$$\mathbf{X} \in \mathbb{R}^{1200 \times 69}, \quad (2)$$

where each row represents one university website and each column represents one of the retained landing-page evaluation attributes.

The final dataset therefore contains approximately:

1,200 universities $\times$ 69 evaluation attributes
--

before the addition of identification fields, labels, derived features, or machine-learning predictions.

5 Data Schema

5.1 Collected Dataset

The scraped university website data were consolidated into a single CSV file, with one row representing one university website. The final collected dataset contains approximately **1,230 university websites** together with identification information, geographical metadata, crawl metadata, and the retained website-evaluation attributes.

Therefore, the final dataset retains **69 website-evaluation attributes**. These attributes are grouped into header and navigation, rankings and recognition, notices and updates, events and media, landing-page content, footer details, visual design, service interaction, technical performance, SEO and metadata, and accessibility and functionality.

Thus, the dataset can be summarised as:

$$1,230 \text{ university websites} \times 69 \text{ evaluation attributes}$$

before including identification and crawl-related metadata.

5.2 Regional Distribution

The university websites were divided among six members of the research team according to geographical region. Approximately 200 universities were assigned to each collector. After consolidation of the scraped records, the dataset contains 1,230 university websites.

Collector	Area of Universities Collected	Universities
Nabil	North America	202
Abhijit	Western & Northern Europe	207
Afifa	Eastern & Southern Europe	201
Faisal	Asia Part 1	200
Mithu	Asia Part 2	212
Mahi	Latin America and Africa	208
Total		1,230

5.3 Raw Scraped Record

Each row of the CSV file corresponds to one university website. In addition to the extracted website attributes, each record contains basic information about the university and the crawling process.

A simplified representation of one scraped record is shown below.

```
{
  "uni_id": 1001,
  "member": "M1",
  "region": "North America",
  "country": "...",
  "name": "...",
  "url": "https://...",
  "final_url": "https://...",
  "http_status": 200,
  "ok": true,
  "page_lang": "en",
```

```

"native_lang": "...",
"switched_to_english": false,
"fetched_at": "...",
"extractor_version": "...",
"load_time_s": ...,
"render_error": null,

"a01_logo": 1,
"a02_primary_nav": 1,
"a03_nav_item_count": 7,
...
"a74_a11y_toggle": 0,
"a75_bookmark": 0
}

```

The identification and crawl-related fields describe the source university and the extraction process, whereas the `aXX_*` fields contain the retained landing-page evaluation attributes.

5.4 Website-Level Dataset Structure

The consolidated CSV is organised into three major groups:

1. university identification and geographical metadata;
2. crawl and extraction metadata; and
3. the 69 retained landing-page evaluation attributes.

Table 3: Schema of the collected university website dataset

Column(s)	Type	Range / Format	Description
Identification and Geographical Metadata			
<code>uni_id</code>	Integer	Unique identifier	Unique identification number assigned to each university.
<code>member</code>	Categorical	M1–M6	Research-team member responsible for collecting the record.
<code>region</code>	Categorical	6 geographical groups	Geographical region assigned during data collection.
<code>country</code>	Categorical	Country name	Country in which the university is located.
<code>name</code>	String	—	Official university name.
<code>url</code>	String	URL	Original university website URL supplied to the scraper.
<code>final_url</code>	String	URL	Final resolved URL after redirection.
Crawl and Extraction Metadata			
<code>http_status</code>	Integer	HTTP status code	HTTP response obtained while accessing the university landing page.
<code>ok</code>	Boolean	True / False	Indicates whether the page was successfully accessed and processed.
<code>page_lang</code>	Categorical	Language code / label	Language detected from the fetched page.

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Column(s)	Type	Range / Format	Description
native_lang	Categorical	Language code / label	Native language associated with the university website where available.
switched_to_english	Boolean / categorical	True / False	Indicates whether an English version of the website was selected during scraping.
fetches_at	Datetime	Timestamp	Time at which the landing page was fetched.
extractor_version	String	Version identifier	Version of the scraper/extractor used during data collection.
load_time_s	Float	Seconds	Observed page-loading time during crawling.
render_error	Nullable string	Error / null	Stores rendering-related error information where applicable.
1. Header and Navigation			
a01_logo	Boolean	0/1	Logo / institutional branding.
a02_primary_nav	Boolean	0/1	Primary navigation menu presence.
a03_nav_item_count	Integer	0–20	Number of top-level navigation items.
a04_search_bar	Boolean	0/1	Search bar presence.
a05_language_toggle	Boolean	0/1	Native-language / English language toggle.
a06_breadcrumb	Boolean	0/1	Breadcrumb or current-location cue.
2. Rankings and Recognition			
a07_qs_badge	Boolean	0/1	QS World Ranking badge or mention.
a08_qs_value	Integer / nullable	1–2000 or null	Displayed QS World Ranking value where available.
a09_national_rank	Boolean	0/1	National or local ranking mention.
a10_webometrics_value	Integer / nullable	1–30000 or null	Displayed Webometrics ranking value where available.
a11_accreditation	Boolean	0/1	Accreditation or certification badge presence.
a12_accred_count	Integer	0–10	Number of accreditation badges displayed.
a13_achievements	Boolean	0/1	Recent achievements or awards section.
a14_stats_block	Boolean	0/1	Impact or highlight statistics block.
a15_stat_item_count	Integer	0–10	Number of highlight-statistic items shown.
3. Notices, Announcements and Updates			
a16_notice_board	Boolean	0/1	Notice-board presence.
a17_notice_timestamp	Boolean	0/1	Indicates whether notice-posting dates are displayed.

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Column(s)	Type	Range / Format	Description
a18_recent_notice_date	Date / nullable	ISO 8601 or null	Most recent notice date detected from the landing page.
a20_news_events	Boolean	0/1	News and events section presence.
a21_calendar_link	Boolean	0/1	Events or academic-calendar link.
a22_admission_notice	Boolean	0/1	Admission announcements or deadlines.
4. Events and Media			
a23_upcoming_events	Boolean	0/1	Upcoming-events section presence.
a24_event_count	Integer	0–20	Number of upcoming events displayed.
a25_event_images	Boolean	0/1	Event images present.
a26_event_captions	Boolean	0/1	Event captions or descriptions present.
a27_event_datetime	Boolean	0/1	Event date and time displayed.
a28_contests	Boolean	0/1	Upcoming contests or competitions announced.
a29_video_content	Boolean	0/1	Video content on the landing page.
a30_image_gallery	Boolean	0/1	Photo or image-gallery presence.
a31_social_feed_embed	Boolean	0/1	Embedded social-media feed presence.
5. Landing-Page Content and Information			
a32_vision_mission	Boolean	0/1	Vision and mission statement.
a33_about_blurb	Boolean	0/1	About or university-overview information.
a34_department_links	Boolean	0/1	Department listing or quick links.
a35_faculty_link	Boolean	0/1	Faculty information or link.
a36_research_highlight	Boolean	0/1	Research or publication highlights.
a37_programs_listing	Boolean	0/1	Academic-program listing.
a38_scholarship	Boolean	0/1	Scholarship information or link.
a39_library_link	Boolean	0/1	Library access or link.
a40_career_link	Boolean	0/1	Career or job-portal link.
a41_alumni_link	Boolean	0/1	Alumni section or link.
a42_faq_link	Boolean	0/1	FAQ link.
a43_contact_link	Boolean	0/1	Contact information or webmail link.
a44_student_portal	Boolean	0/1	Student-portal link.
a45_prospectus	Boolean	0/1	Prospectus or brochure download.
a46_admissions_policy	Boolean	0/1	Admissions policy or procedure link.

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Column(s)	Type	Range / Format	Description
6. Footer Details			
a47_footer_contact	Boolean	0/1	Footer contact details.
a48_footer_sitemap	Boolean	0/1	Footer sitemap link.
a49_copyright_line	Boolean	0/1	Copyright or last-updated line.
a50_social_links	Boolean	0/1	Social-media links.
a51_quick_links	Boolean	0/1	Quick-links block.
7. Visual Design			
a53_contrast_ratio	Float	1–21	Colour contrast ratio based on the WCAG scale.
a54_banner_carousel	Boolean	0/1	Banner or carousel content.
a57_logo_prominence	Boolean	0/1	Noticeable or prominent logo placement.
8. Service Interaction Quality			
a58_live_chat	Boolean	0/1	Live chat or chatbot presence.
a59_feedback_form	Boolean	0/1	Feedback or contact form.
a60_trust_seal	Boolean	0/1	Trust or security seal displayed.
a61_testimonials	Boolean	0/1	Testimonials or student-stories section.
9. Technical Performance			
a62_load_speed_s	Float	Seconds	Landing-page load speed.
a63_mobile_score	Float	0–100	Mobile responsiveness score.
a65_https	Boolean	0/1	HTTPS / SSL presence.
a66_broken_links	Integer	≥ 0	Number of broken links detected on the landing page.
a67_gzip	Boolean	0/1	Page compression such as Gzip enabled.
10. SEO and Metadata			
a69_title_meta	Boolean	0/1	Page title and meta-description presence.
a70_favicon	Boolean	0/1	Favicon presence.
a71_sitemap_robots	Boolean	0/1	Sitemap or robots.txt reference.
11. Accessibility and Functionality			
a72_alt_text_pct	Float	0–100	Percentage of images containing alternative text.
a73_accessible_design	Boolean	0/1	Accessible design across devices.
a74_a11y_toggle	Boolean	0/1	Text-size or contrast accessibility toggle.
a75_bookmark	Boolean	0/1	Bookmark or save-page facility.

5.5 Feature Representation

The retained attributes contain multiple data types. Most landing-page characteristics are presence/absence variables and are therefore represented using binary values:

$$x_i = \begin{cases} 1, & \text{if the corresponding website attribute is detected,} \\ 0, & \text{otherwise.} \end{cases}$$

Count-based attributes, such as the number of navigation items, accreditation badges, events, highlight statistics, and broken links, are stored as integers. Continuous attributes, such as colour contrast ratio, mobile responsiveness, page-load speed, and alt-text coverage, are stored as floating-point values.

The most recent notice date is stored as a temporal value, while ranking information such as QS and Webometrics values may be nullable because such information is not necessarily displayed by every university website.

5.6 Missing and Nullable Values

Not all attributes are applicable or available for every university. In particular, ranking values and notice dates may be unavailable even when the website itself was successfully crawled. Such missing values are retained as null or missing observations during raw data collection instead of automatically assigning them a value of zero.

This distinction is important because:

$$\text{missing value} \neq \text{attribute absent}$$

For example, a missing QS ranking value does not necessarily indicate that the university has no ranking; it only indicates that a ranking value was not successfully extracted from the landing page. Appropriate missing-value handling is therefore applied during preprocessing before model training.

5.7 Sample Dataset Structure

The following example illustrates the organisation of a small subset of the columns. Only selected attributes are shown for readability.

uni_id	region	country	a01_logo	a16_notice_board	a62_load_speed_s	a72_alt_text_pct
1001	North America	—	1	1	—	—
1002	North America	—	1	0	—	—
1003	North America	—	1	1	—	—

The complete dataset contains the identification and crawl metadata together with all 69 retained evaluation attributes.

5.8 Preprocessing for Machine Learning

Before machine-learning experiments, columns that identify a university or describe the scraping process are separated from the actual evaluation features. Fields such as `uni_id`, `name`, `url`, `final_url`, `member`, and other crawl-control information are therefore not treated as website-quality predictors unless explicitly required for analysis.

Binary features retain their 0/1 representation. Numerical count and continuous features retain their numerical values and may be normalised or standardised depending on the selected learning algorithm. Nullable fields are handled using an appropriate missing-value strategy.

Date-valued attributes such as `a18_recent_notice_date` can be transformed into a more useful numerical feature, for example:

$$\text{notice recency} = \text{crawl date} - \text{most recent notice date}.$$

This provides a numerical measure of how recently information was updated without using the excluded notice-update-frequency attribute.

5.9 Modelling Representation

After removing identifiers and crawl-control metadata, the retained website-quality representation consists of 69 evaluation attributes. Consequently, before further feature selection, transformation, or dimensionality reduction, the model input can be represented approximately as

$$\mathbf{X} \in \mathbb{R}^{1230 \times 69},$$

where each row represents one university website and each column corresponds to one retained landing-page evaluation attribute.

The final feature order is fixed before model training so that the same representation is maintained during training, validation, testing, and future inference.

Therefore, the overall data flow can be summarised as:

University URL $\rightarrow$ Landing-page crawl $\rightarrow$ Attribute extraction $\rightarrow$ CSV dataset $\rightarrow$ Preprocessing $\rightarrow \mathbf{X}_{1230 \times 69}$

6 ML Study — Pipeline Design

This section specifies the end-to-end pipeline that converts a list of university URLs into a trained, interpretable website-quality model and a ranked institutional list. The pipeline operates on the dataset defined in the preceding sections, namely

$$\mathbf{X} \in \mathbb{R}^{1230 \times 69},$$

where each row is one university landing page and each column is one retained evaluation attribute. Stages S1–S3 (frame, crawl, parse) and S6 (preprocessing) are specified in the Data Collection, Data Schema, and Preprocessing sections and are referenced here only by their inputs and outputs; the remaining stages are specified in full below.

6.1 Pipeline Overview

The pipeline is organised as ten sequential stages. Each stage emits a versioned artefact so that any downstream result can be traced back to the exact crawl and feature build that produced it.

#	Stage	Operation	Output artefact
1	Frame construction	Region-partitioned university list, official domain verification, duplicate and defunct-domain removal	universities.csv ($n=1,230$)
2	Crawl	Headless-Chrome rendering, 3 runs per site, robots.txt respected, read-only requests	raw HTML, DOM snapshots, screenshots, timing objects
3	Parse	DOM traversal, keyword and link extraction, asset and status checking	page table, link table, asset table
4	Feature build	69 attribute extractors grouped into 11 blocks; multilingual keyword matching; nullable ranking fields preserved	webqual_v1.csv ($1,230 \times 69$)
5	Label build	Rule composite, ordinal grade, expert gold subset	labels.csv (3 label columns)
6	Preprocess	Typed imputation, encoding, log1p , scaling fitted inside CV folds	model-ready design matrix
7	Train	Repeated stratified 5-fold CV with nested inner 3-fold tuning	fitted models, hyper-parameter logs
8	Evaluate	Comparison against baselines and gold labels; block-wise ablation	CV results table, ablation chart
9	Explain	Permutation importance and SHAP; per-university gap analysis	importance report, recommendation sheets
10	Deliver	Ranked institutional list, per-attribute visualisations, predict(url) inference service	final report, released dataset, service endpoint

The complete data flow is therefore:

URL frame $\rightarrow$ crawl $\rightarrow$ parse $\rightarrow \mathbf{X}_{1230 \times 69} \rightarrow$ labels $\rightarrow$ CV training $\rightarrow$ evaluation $\rightarrow$ explanation $\rightarrow$ ranking.

6.2 Feature Blocks

The 69 retained attributes are partitioned into eleven semantically coherent blocks. This partition is the unit of analysis for the ablation study in Section 6.7 and for the interpretation of block-level SHAP contributions.

#	Feature block	Attributes	Nature of the signal
B1	Header & navigation	6	Wayfinding and structural usability
B2	Rankings & recognition	9	Externally conferred prestige signals
B3	Notices & updates	6	Content currency and maintenance
B4	Events & media	9	Engagement and multimedia richness
B5	Landing-page content	15	Information completeness
B6	Footer details	5	Secondary navigation and credibility
B7	Visual design	3	Appearance and readability
B8	Service interaction	4	Two-way interaction and trust
B9	Technical performance	5	Speed, security, reliability
B10	SEO & metadata	3	Discoverability
B11	Accessibility & functionality	4	Inclusive access
Total		69	

6.3 Label Construction

Because no externally published ground-truth quality score exists for 1,230 university websites, three label variants are constructed and every reported metric is tagged with the variant it was computed against.

- L1. `label_rule_score`** (weak, all 1,230 rows). A composite score on $[0, 100]$ formed by block-normalised aggregation of the 69 attributes. Binary attributes contribute their block-normalised presence, counts and continuous measurements are min-max scaled within block after winsorisation, and each block contributes according to a declared weight vector. This label is cheap and complete but is an arithmetic function of the features, so it is used only for pre-training diagnostics and never as the headline evaluation target.
- L2. `label_grade`** (5-level ordinal, all rows). Obtained by cutting `label_rule_score` at its quintile boundaries, yielding grades Poor, Fair, Good, Very Good, and Excellent with approximately 246 sites each. Balanced strata make stratified cross-validation straightforward and permit the use of ordinal-aware losses.
- L3. `label_expert_score`** (gold, stratified subset). A human-rated subset of 180 sites, 30 per region, sampled across the full range of `label_rule_score` so that the subset is not concentrated in the middle of the distribution. Each site is independently rated on $[0, 100]$ by three trained raters using a fixed rubric covering information completeness, navigability, currency, appearance, and technical quality. Inter-rater reliability is reported as Krippendorff’s α and as the mean pairwise Spearman ρ ; the rubric is revised and the batch re-rated if $\alpha < 0.67$. The mean of the three ratings is the gold score.

Leakage control. Two safeguards are applied. First, the displayed-ranking attributes `a08_qs_value` and `a10_webometrics_value` measure institutional prestige rather than website quality, and a model given them can rank universities well while learning nothing about their websites; they are therefore excluded from the primary feature set and retained only for a secondary prestige-controlled analysis. Second, any externally sourced Webometrics rank is held out entirely from training and used only as an independent construct-validity check on the final ranking. Human raters are blind to `label_rule_score` and to institutional identity where the rubric permits.

6.4 Modelling Tracks

Track A — ordinal classification (5 grades). Random Forest, LightGBM/XGBoost, SVM-RBF, MLP, and an ordinal-aware head (ordinal logistic or CORAL/CORN). Metrics: accuracy, macro-F1, and *quadratic weighted kappa*, which penalises an Excellent $\leftrightarrow$ Poor confusion far more heavily than an adjacent-grade confusion, together with the full confusion matrix.

Track B — regression (0–100). Random Forest, gradient-boosting, and MLP regressors, with Ridge regression retained as an interpretable floor. Metrics: RMSE, MAE, R^2 , and the Spearman ρ between the induced ranking and the expert ranking. Since the deliverable is a rank list, rank correlation is treated as the primary criterion and absolute error as secondary.

Track C — learning to rank (optional). LambdaMART evaluated by NDCG@10 and NDCG@50, trained on the expert subset with region used as the query grouping so that the model learns within-region orderings rather than a single global list.

Primary model. LightGBM on the tabular feature set is the recommended primary model: it handles the mixed binary/count/continuous representation and the nullable fields natively, is robust at this sample size, and yields per-feature and per-block importances directly. An MLP is trained in parallel purely as a comparison against Biyyapu et al. (2023), whose architecture it mirrors.

6.5 Validation Protocol

All models are evaluated with repeated stratified 5-fold cross-validation (5 repetitions, different seeds), stratified jointly on `label_grade` and `region` so that no fold is dominated by a single geography. Hyper-parameters are selected by a nested inner 3-fold search executed strictly within each training fold, and all scalers, imputers, and encoders are fitted inside the fold to prevent information from the validation partition reaching the model. Results are reported as mean $\pm$ standard deviation across the 25 fold-repetition pairs.

Two additional evaluations are performed. A *leave-one-region-out* split tests whether a model trained on five regions generalises to an unseen sixth, which is the practical test of whether the learned notion of quality is region-specific. A *gold-subset evaluation* scores the model trained on weak labels against `label_expert_score` on the 180 rated sites; this is the headline number, because it measures agreement with human perception rather than with the scoring rule the model was fitted to.

6.6 Baselines

A learned model is reported as successful only if it beats all three of the following:

- **Majority class** (Track A) and **mean predictor** (Track B), establishing the trivial floor.
- **The rule-based composite**, i.e. `label_rule_score` evaluated directly against `label_expert_score`. This is the baseline the project exists to displace: if the hand-weighted formula already agrees with human raters as well as the model does, the machine-learning contribution is null.
- **Single-block predictors**, in particular the content block (B5) alone and the technical block (B9) alone. A 69-feature model that cannot beat a 15-feature content-only model does not justify its collection cost.

6.7 Ablation by Feature Block

Models are trained incrementally over the eleven blocks in the order B5 (content) $\rightarrow$ +B1 $\rightarrow$ +B3 $\rightarrow$ +B9 $\rightarrow$ +B11 $\rightarrow$ +B10 $\rightarrow$ +B7 $\rightarrow$ +B8 $\rightarrow$ +B4 $\rightarrow$ +B6 $\rightarrow$ +B2, and each configuration is scored against `label_expert_score`. A complementary leave-one-block-out pass quantifies the marginal contribution of each block to the full model.

The headline analytical result is the comparison between the SHAP-derived empirical block weights and the weights assigned a priori in the rule-based composite. Where the two disagree, the learned weights constitute direct evidence that hand-set scoring rules misallocate importance across quality dimensions — which is the central claim of this work.

6.8 Risks

Risk	Impact	Mitigation
Label leakage: the rule label is an arithmetic function of the features	Reported accuracy is circular and meaningless	Expert gold score is the primary evaluation label; every metric is tagged with its label variant; rule label used only for diagnostics
Prestige leakage via displayed QS/Webometrics values	Model ranks universities, not websites	Ranking-value attributes excluded from the primary feature set; external ranks held out for construct validity only
Inconsistent extraction across six collectors	Systematic between-region offsets mistaken for quality differences	Single shared extractor version; 60-site overlap sample re-crawled by a second collector; inter-collector agreement reported per attribute
Network and geographic variance in timing measurements	Spurious performance gaps favouring nearby regions	Fixed measurement host, median of three runs, per-run control-URL covariate, region included as a covariate in performance analyses
Non-English and JS-heavy sites under-extracted	Content attributes systematically under-counted	Headless-Chrome rendering, multilingual keyword and embedding matching, manual spot-check of 60 sites stratified by language
Regional bias in the learned quality notion	Model encodes wealth or region rather than quality	Leave-one-region-out evaluation; region-stratified folds; per-region error reporting
Expert-rating shortfall or low agreement	No usable gold labels	Recruit raters early; rubric pilot before the main batch; re-rate if Krippendorff's $\alpha < 0.67$
Nullable ranking and date fields imputed as zero	Absence of a displayed value confused with a poor value	Typed missingness with explicit indicator columns, as specified in the Preprocessing section

7 Work Plan

Week	Milestone	Output
1	Freeze UGC frame, verify domains	<code>universities.csv</code> (n≈180)
2–3	Crawler + parser, pilot on 20 sites	working crawler, pilot JSON
4	Full crawl, runs 1–3	complete raw dataset
5	Feature pipeline, detector validation	<code>webqual_bd_v1.csv</code> , precision/recall report
5–7	Survey deployment	≥350 responses, gold labels
6–8	Training, tuning, ablation	CV results table, ablation chart
9	Interpretability, dashboard	SHAP report, recommendation sheets
10	Final report, dataset release	report + open dataset

8 Summary

Item	Content
Data source	Census crawl of all UGC-listed Bangladeshi universities (175–185 in frame, 165–180 usable), 3 runs each, plus a 30–40 site student survey
Dataset sizes	~170 website rows × ~30 features; ~2,000 pages; ~70,000 links/assets; 4,250 content judgements; 350–500 survey responses
Feature groups	Content (25), completeness ratios (7+), performance durations (12), load time (1); optional: accessibility, SEO, appearance, bilingual, Webometrics
Labels	<code>label_rule_score</code> (weak), <code>label_grade</code> (5-level ordinal), <code>label_survey_score</code> (gold)
Models	LightGBM (primary), RF, MLP, SVM-RBF, ordinal classifier; NLP branch: multilingual Sentence-BERT, TF-IDF + linear SVM
Evaluation	Repeated stratified 5-fold CV; accuracy / macro-F1 / quadratic weighted kappa; RMSE / MAE / R^2 / Spearman ρ ; ablation by block
Novelty	First national-scale <i>learned</i> quality model for Bangladeshi university websites, validated against human perception rather than against its own scoring rule

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